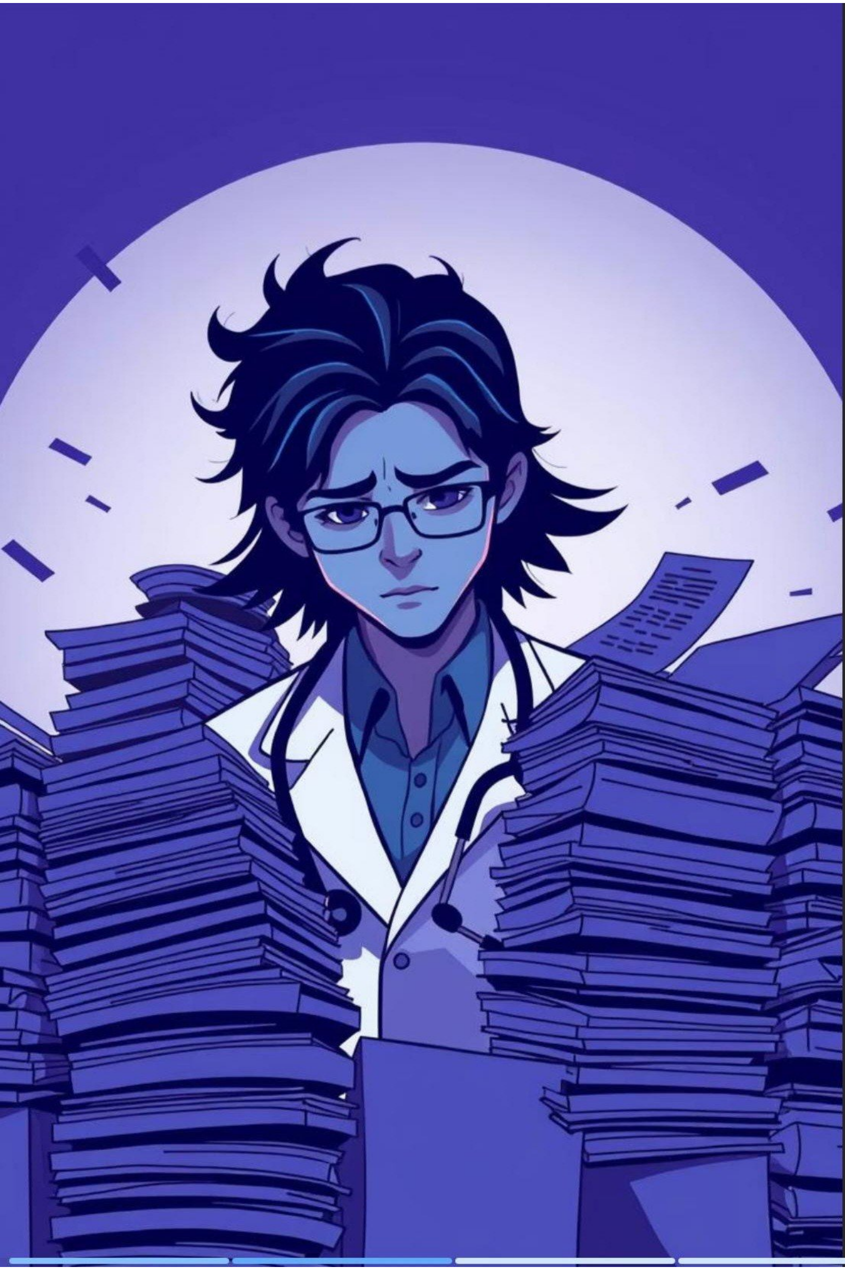
MediBot – Medical Report Chatbot

AI-powered Medical PDF Report Interpreter



The Challenge: Unstructured Medical Data

Healthcare relies on vast amounts of information, often locked within unstructured PDF documents. Extracting and interpreting this data manually is slow, error-prone, and inefficient.

> "Processing unstructured medical text from PDFs presents significant challenges, demanding real-time, contextually accurate AI solutions for effective healthcare data interpretation."

MediBot: System Overview & Objectives

MediBot is an integrated AI system designed for precise extraction, intelligent indexing, and accurate interpretation of complex medical documents.

High-Accuracy Parsing

Ensure precise extraction of clinical details from diverse PDF formats.

Rapid Query Resolution

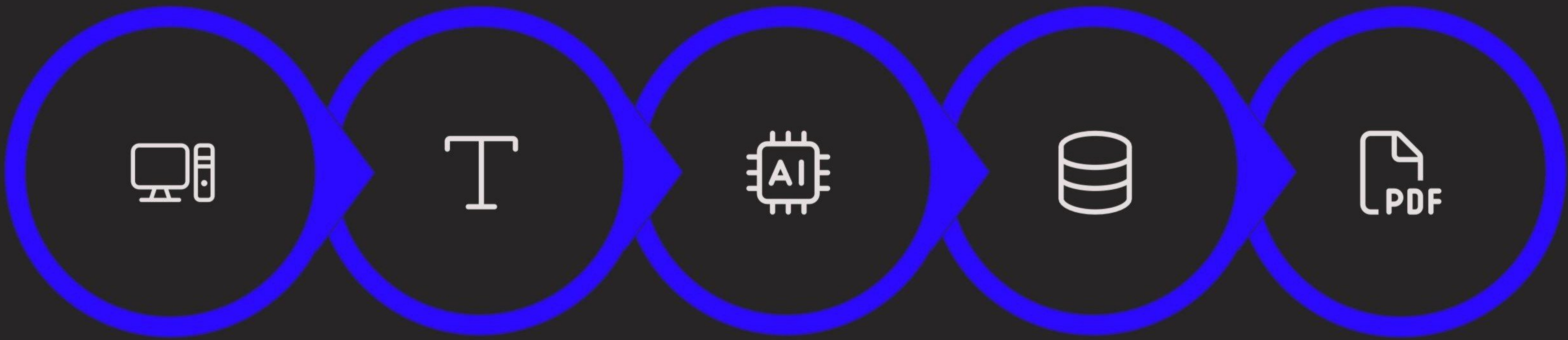
Enable real-time, context-aware responses to user queries.

Robust Local Deployment

Provide a secure, efficient solution for on-premise operation.

Technology Stack

MediBot leverages a robust open-source stack for secure, efficient, and accurate medical document interpretation.



Streamlit Frontend

SentenceTransformers NLP

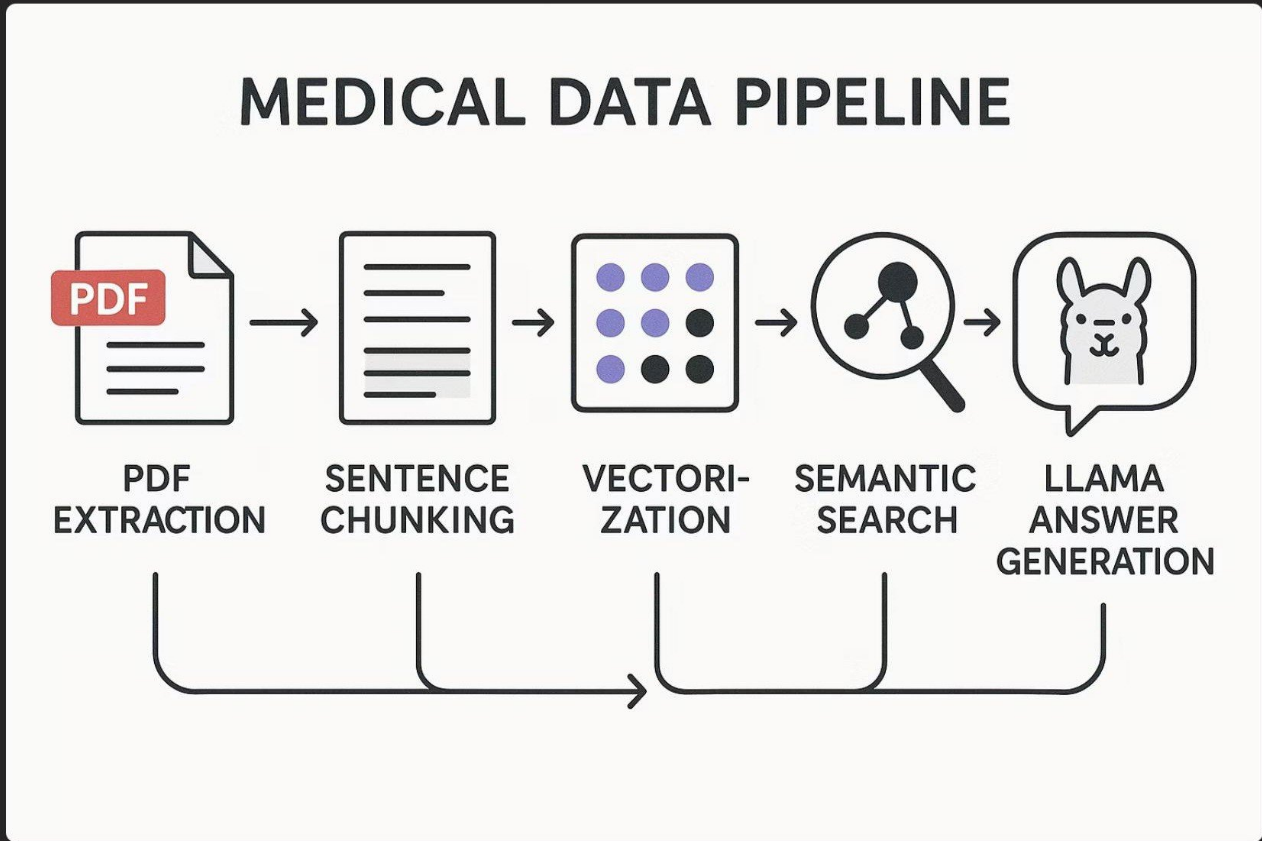
LLaMA 3 with Ollama

ChromaDB Vector DB

PyPDF2/pdfplumber

Architecture & Data Pipeline

MediBot's pipeline ensures seamless data flow from PDF upload to natural language response, prioritizing efficiency and accuracy.



Algorithmic & Model Deep Dive

SentenceTransformers

- Fine-tuned on biomedical datasets (e.g., PubMed, MIMIC-III).
- Optimized for semantic similarity in clinical contexts.

LLaMA 3 & Prompt Engineering

- Context-specific prompts designed for diagnostic, treatment, and prognostic inquiries.
- Few-shot learning examples improve response relevance.

Context window management in ChromaDB uses hybrid retrieval (BM25 + vector search) for highly relevant context extraction.

Security & Privacy

Ensuring the confidentiality and integrity of Protected Health Information (PHI) is paramount for MediBot.

Local Inference

LLaMA 3 runs on-premises via Ollama, ensuring no PHI leaves the local environment.

Data Isolation

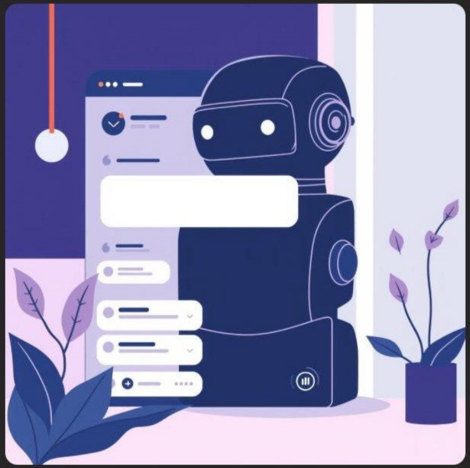
Strict data partitioning and access controls prevent unauthorized data exposure.

PHI Anonymization

Best practices for de-identification applied to sensitive data where necessary.

Demonstration & Performance

Streamlit UI Walkthrough



Interactive demonstration of PDF upload, query input, and real-time response generation.

Benchmarking Results

MediBot demonstrates impressive performance across key metrics, ensuring rapid and accurate insights.

Technical Challenges & Solutions

Developing MediBot required overcoming several complex technical hurdles to ensure robust performance.

1

Complex Table/Figure Extraction

Solution: Advanced rule-based parsing combined with vision models for structural understanding.

2

Ambiguous Medical Terminology

Solution: Domain-specific embeddings and knowledge graph integration for disambiguation.

3

Rapid Semantic Search

Solution: Optimized ChromaDB indexing strategies and quantized embeddings for efficiency.

Conclusion: Transforming Medical Data Accessibility

MediBot represents a significant leap forward in making complex medical reports accessible and actionable.

- **Technical Feats:** Seamless integration of leading AI/NLP models, robust local deployment, and efficient data pipelines.
- **Impact:** Reduces manual data interpretation burden, accelerates clinical insights, and enhances healthcare decision-making.

MediBot is poised to revolutionize how we interact with medical information, empowering healthcare professionals with immediate, accurate data.